MAHATMA GANDHI UNIVERSITY

BCA (HONOURS) FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG5DSEBCA304 -Digital Image Processing

Model Question Paper

Max. Marks: 70 **Time 2 Hours**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1	Describe brightness adaptation.	[U][CO 1]
2	Define digital image processing. Compute the space requirement of a 250X250 color image.	[U][CO 1]
3	Explain the role of basic set operations in image processing	[An][CO 3]
4	Explain the concepts of dilation operation.	[U][CO 3]
5	How does impulse noise affect an image,.	[R][CO 4]

(5*2=10 Marks)

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6	Compare different color models	[An][CO 1]
7	Explain the different components of Digital image processing system.	[U][CO 1]
8	Describe the fundamental steps in digital image processing.	[U][CO 1]
9	Explain basic intensity transformation functions.	[U][CO 2]
10	Analyse the effect of Histogram matching on the contrast and brightness of an image	[An][CO 2]
11	Explain the use of closing and opening operation in noise removal.	[An][CO 3]
12	Compare horizontal and vertical edge detection filters.	[An][CO 4]

(6*5=30 Marks)

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13	Compare smoothing and sharpening spatial filters	[An][CO 2]
14	Explain Hit-or-miss transformation with suitable example, How it is used for pattern recognition	[An][CO 3]
15	Compare different region based segmentation methods,	[An][CO 4]

(2*15=30 Marks)